CF terminology & relevant concepts

Digital evidence

- **data stored or transmitted in digital form that can be used in court**
- **the cornerstone of any digital forensics investigation**
  - data can be viewed at different levels of abstraction, requires interpretation, are fragile, may be voluminous
  - Often is difficult to discover connection data <-> reality
- **requires technical understanding of different possible types (files, emails, logs, metadata) and the legal requirements for collecting and preserving it**
- **knowledge of file systems, network protocols, and encryption methods essential as well**

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Chain of custody

- **documented and "unbroken" process of handling evidence from the time it is collected until it is presented in court**
- **ensures that digital evidence has not been tampered with, altered, or accessed by unauthorized parties**
  - critical for the evidence to remain legally admissible in investigations and trials
- **requires knowledge about how to document evidence collection, storage, and access**
  - involves knowledge of logging procedures, secure storage solutions, and legal protocols.

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Data acquisition

- **the process of collecting digital evidence from devices without altering or damaging the original data.**
- **requires knowledge of disk imaging and live data capture**
- **requires expertise in forensics acquisition and analysis tools, like *FTK Imager, EnCase* (even *dd*)**
- **knowledge of file systems, write-blockers, and hashing is crucial for ensuring integrity**

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Hashing

- **the process of converting data into a fixed-length string of bits, which represents the data uniquely**
- **to ensure the integrity of digital evidence. A hash value can verify that a file has not been altered during the investigation**
- **requires understanding of hashing algorithms (strengths and weaknesses, e.g. MD5 collision) and formats (*hex, base64*)**
- **requires expertise in hashing tools**
  - *sha256sum, hashdeep*
  - *FTK imager, Autopsy*
- **have to be used any time an evidence is "managed" (copied, moved)**

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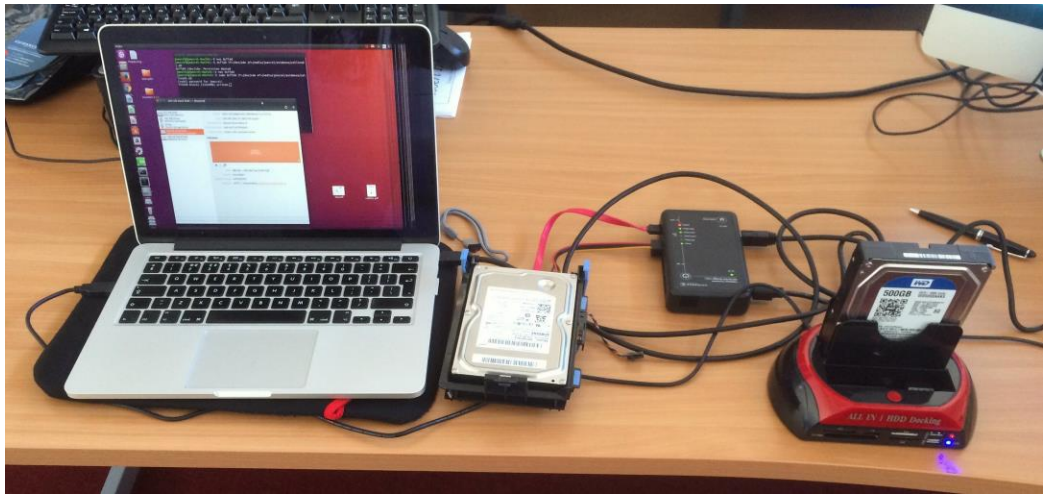
Write Blocker

- hardware or software tool used to prevent any data from being written to a storage device during analysis, preserving the original data content
- requires understanding of how write-blocking devices work and how they can be implemented in forensic procedures
- essential for legally defensible acquisitions



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Forensic image

- **A bit-by-bit copy of digital media, including deleted files and data in slack space, which is an exact replica of the original device**
- **Goal: to preserve the original evidence.**
  - **avoiding modification of the original data**
- **requires understanding of mechanisms to copy information in digital devices (file system knowledge and behavior)**
- **requires familiarity with bit-by-bit copy tools (a.k.a. forensic imaging tools)**
  - *dd (and command line evolutions)*
  - *FTK imager, Encase, Guymager, ...*
- ***have to be used any time an evidence is "managed" (copied, moved)***

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Scenarios

Typical computer forensics scenario

More on Prof. Vaciago' lessons

- **internet abuse from employee**
- **computer-aided frauds**
- **data unauthorized manipulation**
  - data theft
  - data disclosure
- **computer/network damage assessment**
- **…and any time digital evidences may be involved in an incident**

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